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### IMAGE PROCESSING DEVICE, IMAGE PROCESSING METHOD, AND PROGRAM

#### Abstract

The present disclosure relates to an image processing device, an image processing method, and a program that are capable of generating a 3D model with high accuracy even when occlusion has occurred.

The image processing device includes: a bone information estimation unit that estimates, based on captured images at a predetermined time captured by a plurality of imaging devices, bone information of a person appearing in the captured images; and a 3D model estimation unit that, when occlusion has occurred on the person appearing in the captured images at the predetermined time, estimates a 3D model of the person based on the bone information at the predetermined time. The technology of the present disclosure can be applied to, for example, an image processing device that distributes free viewpoint video.

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## Background/Summary

### TECHNICAL FIELD

[0001] The present disclosure relates to an image processing device, an image processing method, and a program, and more particularly, relates to an image processing device, an image processing method, and a program that are capable of generating a 3D model with high accuracy even when occlusion has occurred.

### BACKGROUND ART

[0002] There is a technology that provides a free viewpoint video by generating a 3D model of a subject from moving images captured from a plurality of viewpoints, and generating a virtual viewpoint video of the 3D model according to any viewpoint (virtual viewpoint) (e.g., see PTL 1). This technology is also called volumetric capture. Using this volumetric capture technology, 3D streaming is being developed that captures a sports game from a plurality of viewpoints and distributes a 3D model of each player in the game to users, which are viewers, so that each user is allowed to freely change the viewpoint while watching the game.

[0003] To capture human movements such as those in sports in 3D with high accuracy, there is a technology that attempts to improve the accuracy of the 3D model shape by breaking down the human body shape (object) in a free viewpoint video content into parts and performing mesh tracking on each part in the time direction (e.g., see PTL 2).

[0004] A rigged human body template model called a skinned multi-person linear model (SMPL) is known, which includes bone information (rigs) such as bones and joints and can accurately represent various human body shapes (e.g., see NPL 1).

### CITATION LIST

#### Patent Literature

##### PTL 1

[0005] WO 2018/150933

##### PTL 2

[0006] Japanese Translation of PCT Application No. 2020-518080

#### Non Patent Literature

##### NPL 1

[0007] SMPL A Skinned Multi-Person Linear Model, <https://smpl.is.tue.mpg.de>

### SUMMARY

#### Technical Problem

[0008] In a volumetric image capture environment such as sports game broadcast as described above, occlusion can inevitably occur due to limitations on the arrangement and number of cameras, the number of players for which 3D models are to be generated, and so on.

[0009] The present disclosure has been made in view of such circumstances, and aims at making it possible to generate a 3D model with high accuracy even when occlusion has occurred.

#### Solution to Problem

[0010] An image processing device according to one aspect of the present disclosure includes: a bone information estimation unit that estimates, based on captured images at a predetermined time captured by a plurality of imaging devices, bone information of a person appearing in the captured images; and a 3D model estimation unit that, when occlusion has occurred on the person appearing

in the captured images at the predetermined time, estimates a 3D model of the person based on the bone information at the predetermined time.

[0011] An image processing method according to one aspect of the present disclosure includes: by an image processing device, estimating, based on captured images at a predetermined time captured by a plurality of imaging devices, bone information of a person appearing in the captured images; and estimating, when occlusion has occurred on the person appearing in the captured images at the predetermined time, a 3D model of the person based on the bone information at the predetermined time.

[0012] A program according to one aspect of the present disclosure is for causing a computer to execute processing of: estimating, based on captured images at a predetermined time captured by a plurality of imaging devices, bone information of a person appearing in the captured images; and estimating, when occlusion has occurred on the person appearing in the captured images at the predetermined time, a 3D model of the person based on the bone information at the predetermined time.

[0013] In one aspect of the present disclosure, based on captured images captured at a predetermined time by a plurality of imaging devices, bone information of a person appearing in the images is estimated; and when occlusion has occurred on the person appearing in the captured images at the predetermined time, a 3D model of the person is estimated based on the bone information at the predetermined time.

[0014] The image processing device according to one aspect of the present disclosure can be realized by causing a computer to execute a program. The program to be executed by the computer can be provided by transmitting through a transmission medium or by recording on a recording medium.

[0015] The image processing device may be an independent device or an internal block constituting one device.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

[0016] FIG. 1 is a block diagram of an image processing system to which the present technology is applied.

[0017] FIG. 2 is a diagram illustrating an example of the arrangement of an imaging device that captures an image of a subject.

[0018] FIG. 3 illustrates a flow of processing in the image processing system of FIG. 1.

[0019] FIG. 4 is a diagram illustrating a problem in occlusion.

[0020] FIG. 5 is a diagram illustrating an example of a captured image in a sports scene.

[0021] FIG. 6 is a diagram illustrating an example of a captured image in a sports scene.

[0022] FIG. 7 is a diagram illustrating an example of a captured image in a sports scene.

[0023] FIG. 8 is a diagram illustrating an example of a captured image in a sports scene.

[0024] FIG. 9 is a diagram illustrating an example of a captured image in a sports scene.

[0025] FIG. 10 is a block diagram illustrating a detailed configuration example of a 3D model generation unit.

[0026] FIG. 11 is a diagram illustrating the generation of an unrigged 3D model.

[0027] FIG. 12 is a diagram illustrating the generation of the unrigged 3D model.

[0028] FIG. 13 is a diagram illustrating the generation of the unrigged 3D model.

[0029] FIG. 14 is a diagram illustrating the generation of the unrigged 3D model.

[0030] FIG. 15 is a diagram illustrating the generation of the unrigged 3D model.

[0031] FIG. 16 is a diagram illustrating data of the unrigged 3D model.

[0032] FIG. 17 is a diagram illustrating the generation of a rigged 3D model.

[0033] FIG. **18** is a diagram illustrating the generation of the rigged 3D model by a rigged 3D model estimation unit.

[0034] FIG. **19** is a flowchart illustrating first 3D moving image generation processing.

[0035] FIG. **20** is a flowchart illustrating second 3D moving image generation processing.

[0036] FIG. **21** is a block diagram illustrating a configuration example of an embodiment of a computer to which the technology of the present disclosure is applied.

#### Description of Embodiments

[0037] Modes for embodying the technology of the present disclosure (hereinafter referred to as “embodiments”) will be described below with reference to the accompanying drawings. In the present specification and the drawings, components having substantially the same functional configuration will be denoted by the same reference numerals, and thus repeated descriptions thereof will be omitted. The description will be made in the following order. [0038] 1. Embodiment of Image Processing System [0039] 2. Flow of Processing in Image Processing System [0040] 3. Problem in Occlusion in Sports Scenes [0041] 4. Detailed Configuration Block Diagram of 3D Model Generation Unit [0042] 5. First 3D Moving Image Generation Processing [0043] 6. Second 3D Moving Image Generation Processing [0044] 7. Conclusion [0045] 8. Computer Configuration Example [0046] 9. Application Examples

#### 1. Embodiment of Image Processing System

[0047] First, an overview of an image processing system to which the present technology is applied will be described with reference to FIGS. **1** to **3**.

[0048] FIG. **1** is a block diagram of the image processing system to which the present technology is applied.

[0049] The image processing system **1** in FIG. **1** is a distribution system that captures images of a person, which is a subject, from a plurality of viewpoints to generate a 3D model by using the volumetric capture technology, and distributes the model to users, which are viewers, so that each user is allowed to freely change the viewpoint to view the subject.

[0050] The image processing system **1** includes a data acquisition unit **11**, a 3D model generation unit **12**, a formatting unit **13**, a transmission unit **14**, a reception unit **15**, a decoding unit **16**, a rendering unit **17**, and a display unit **18**.

[0051] The data acquisition unit **11** acquires image data for generating a 3D model of a subject. The data acquisition unit **11** acquires, for example, a) as image data, a plurality of viewpoint images (referred to as captured images or simply as images) captured by a plurality of imaging devices **41** (hereinafter referred to as cameras **41**) (for a plurality of viewpoints) arranged to surround a subject **31** as illustrated in FIG. **2**. In this case, the plurality of viewpoint images are preferably images captured synchronously by the plurality of cameras **41**. The data acquisition unit **11** may also acquire, for example, b) as image data, a plurality of viewpoint images of a subject **31** captured by one camera **41** from a plurality of viewpoints. The data acquisition unit **11** may also acquire, for example, c) as image data, one captured image of a subject **31**. In this case, the 3D model generation unit **12**, which will be described below, generates a 3D model by using, for example, machine learning.

[0052] The data acquisition unit **11** acquires internal parameters and external parameters, which are camera parameters corresponding to the position (installation position) and attitude of each camera **41**, from outside. Alternatively, the data acquisition unit **11** may perform calibration based on the image data to generate internal parameters and external parameters of each camera **41**. The data acquisition unit **11** may acquire, for example, a plurality of pieces of depth information indicating distances from a plurality of viewpoints to the subject **31**.

[0053] The 3D model generation unit **12** generates, based on image data obtained by capturing images of the subject **31** from a plurality of viewpoints, a model (3D model) having three-dimensional information of the subject. The 3D model generation unit **12** generates a 3D model of the subject by using, for example, a so-called a visual hull to sharpen the three-dimensional shape

of the subject using images from a plurality of viewpoints (e.g., silhouette images from a plurality of viewpoints). In this case, the 3D model generation unit **12** can further deform the 3D model generated using the visual hull with high accuracy by using a plurality of pieces of depth information indicating distances from the plurality of viewpoints to the subject **31**. For example, the 3D model generation unit **12** may generate a 3D model of a subject **31** from one captured image of the subject **31**. The 3D model generated by the 3D model generation unit **12** can be said to be a moving image of the 3D model when generated on a per-frame basis in a time series. Since the 3D model is generated using the image(s) captured by the camera(s) **41**, it can also be said to be a live-action 3D model. The 3D model can express shape information that represents the surface shape of a subject in the form of mesh data, for example, called a polygon mesh, which is expressed by connections between vertices. The method of expressing the 3D model is not limited to the above examples, and the 3D model may be described by a so-called point cloud expression method, which uses position information of points for expression.

[0054] In association with such 3D shape data, texture data, which is color information data, is generated. The texture data includes view independent texture data in which color information remains the same regardless of the viewing direction, and view dependent texture data in which color information changes depending on the viewing direction.

[0055] The formatting unit **13** converts data of the 3D model generated by the 3D model generation unit **12** into a format suitable for transmission and storage. For example, the 3D model generated by the 3D model generation unit **12** may be converted into a plurality of two-dimensional images by perspective projection from a plurality of directions. In this case, the 3D model may be used to generate depth information that is two-dimensional depth images from a plurality of viewpoints. The depth information as the two-dimensional images and color information are compressed and output to the transmission unit **14**. The depth information and the color information may be transmitted as one image in which they are arranged side by side, or may be transmitted as two separate images. In this case, since they are in the form of two-dimensional image data, they can also be compressed using a two-dimensional compression technology such as advanced video coding (AVC).

[0056] For example, the 3D model data may be converted into a point cloud format. It may be output to the transmission unit **14** as three-dimensional data. In this case, it is possible to use, for example, a three-dimensional compression technology of Geometry-based-Approach discussed in MPEG.

[0057] The transmission unit **14** transmits the transmission data formed by the formatting unit **13** to the reception unit **15**. The transmission unit **14** performs a series of processing of the data acquisition unit **11**, the 3D model generation unit **12**, and the formatting unit **13** offline and then transmits the transmission data to the reception unit **15**. Further, the transmission unit **14** may transmit the transmission data generated from the series of processing described above to the reception unit **15** in real time.

[0058] The reception unit **15** receives the transmission data transmitted from the transmission unit **14**.

[0059] The decoding unit **16** performs decoding processing on the transmission data received by the reception unit **15**, and decodes the received transmission data into 3D model data (shape and texture data) required for display.

[0060] The rendering unit **17** performs rendering using the 3D model data decoded by the decoding unit **16**. For example, texture mapping is performed in which a mesh of the 3D model is projected from the viewpoint of a camera that draws it, and a texture that represents color and pattern is applied. This rendering can be viewed from any free viewpoint set regardless of the camera position at the time of capturing.

[0061] The rendering unit **17** performs texture mapping to apply a texture representing the color, pattern, and quality feel of a mesh depending on the position of the mesh of the 3D model, for

example. Texture mapping includes a method called view dependent that considers the user's viewing viewpoint, and a method called view independent that does not consider the user's viewing viewpoint. The view dependent method has the advantage of achieving higher quality rendering than the view independent method because it changes the texture applied to the 3D model depending on the position of the viewing viewpoint. On the other hand, the view independent method has the advantage of reducing the amount of processing compared to the view dependent method because it does not consider the position of the viewing viewpoint. The data of the viewing viewpoint is detected as the user's viewing position (region of interest) by a display device, and is input from the display device to the rendering unit **17**. The rendering unit **17** may also employ billboard rendering, which renders an object so that the object maintains a vertical posture with respect to the viewing viewpoint. For example, when rendering a plurality of objects, objects that are of low interest to the viewer may be rendered as billboards, while the other objects may be rendered using another rendering scheme.

[0062] The display unit **18** displays the result of rendering by the rendering unit **17** on a display of the display device. The display device may be a 2D monitor or a 3D monitor such as a head-mounted display, a spatial display, a mobile phone, a television set, or a PC.

[0063] FIG. **1** illustrates, in the image processing system **1**, a series of flows from the data acquisition unit **11** that acquires captured images, which are materials for generating content, to the display unit **18** that controls the display device on which the user views. However, this does not mean that all functional blocks are necessary to implement the present technology, and the present technology may be implemented for each functional block or a combination of a plurality of functional blocks. For example, in FIG. **1**, the transmission unit **14** and the reception unit **15** are provided to indicate a series of flows from a side that creates content to a side that views the content through distribution of content data, but the flow from content creation to viewing may be implemented by a single image processing device (e.g., a personal computer). In this case, the formatting unit **13**, the transmission unit **14**, the reception unit **15**, or the decoding unit **16** can be eliminated from the image processing device.

[0064] In implementing the image processing system **1**, the same implementer may implement everything, or different implementers may implement each functional block. As an example, operator A generates 3D content through the data acquisition unit **11**, the 3D model generation unit **12**, and the formatting unit **13**. The 3D content may then be distributed through the transmission unit **14** (platform) of operator B, and a display device of operator C may receive, render, and control to display the 3D content.

[0065] Each functional block can be implemented on the cloud. For example, the rendering unit **17** may be implemented in the display device or may be implemented in a server. In that case, information is exchanged between the display device and the server.

[0066] In FIG. **1**, the data acquisition unit **11**, the 3D model generation unit **12**, the formatting unit **13**, the transmission unit **14**, the reception unit **15**, the decoding unit **16**, the rendering unit **17**, and the display unit **18** are collectively described as the image processing system **1**. However, in the present specification, the image processing system **1** will be referred to as an image processing system if two or more functional blocks are related to one another, and for example, the data acquisition unit **11**, the 3D model generation unit **12**, the formatting unit **13**, the transmission unit **14**, the reception unit **15**, the decoding unit **16**, and the rendering unit **17**, excluding the display unit **18**, can be collectively referred to as the image processing system **1**.

## 2. Processing Flow in Image Processing System

[0067] A flow of processing in the image processing system **1** will be described with reference to a flowchart of FIG. **3**.

[0068] When the processing starts, the data acquisition unit **11** acquires image data for generating a 3D model of the subject **31** in step **S11**. In step **S12**, the 3D model generation unit **12** generates a model having three-dimensional information of the subject **31** based on the image data for

generating the 3D model of the subject **31**. In step **S13**, the formatting unit **13** encodes a shape and texture data of the 3D model generated by the 3D model generation unit **12** into a format suitable for transmission or storage. In step **S14**, the transmission unit **14** transmits the encoded data, and in step **S15**, the reception unit **15** receives the transmitted data. In step **S16**, the decoding unit **16** performs decoding processing to perform conversion into a shape and texture data required for display. In step **S17**, the rendering unit **17** performs rendering using the shape and texture data. In step **S18**, the display unit **18** displays the result of rendering. When the processing of step **S18** ends, the processing of the image processing system ends.

### 3. Problem in Occlusion in Sports Scenes

[0069] As an example of an application of the image processing system **1** described above, a case will be considered in which the image processing system **1** is applied to broadcasting a sports game. For example, as illustrated in FIG. **4**, a plurality of cameras **41** indicated by circles are installed at a basketball game venue. The plurality of cameras **41** are installed to surround the court from various positions and directions, such as close to the court and at a distance to capture an image of the entire court. In the example of FIG. **4**, the numbers illustrated in the circles representing the respective cameras **41** are camera numbers for identifying the plurality of cameras **41**, and in the example of FIG. **4**, a total of 22 cameras **41** are installed, but the arrangement and number of cameras **41** are merely an example and are not limited to this example.

[0070] In a volumetric image capture environment such as sports game broadcast, occlusion can inevitably occur due to limitations on the arrangement and number of cameras **41**, the number of players for which 3D models are to be generated, and so on.

[0071] For example, images **P1** to **P5** in FIGS. **5** to **9** provide examples of images captured during a game at the same time by five cameras **41** installed at the game venue.

[0072] In the images **P1** to **P5** illustrated in FIGS. **5** to **9**, the people for which 3D models are to be generated are, for example, players and a referee in the game. Of the players and the referee appearing in the images **P1** to **P5**, the player enclosed by a rectangular frame is set as a target model **TG**, which is an object for which a 3D model is to be generated.

[0073] In the image **P1** of FIG. **5** captured by a first camera **41**, the target model **TG** partially overlaps with another player **PL1**, resulting in occlusion.

[0074] In the image **P2** of FIG. **6** captured by a second camera **41**, the target model **TG** does not overlap with any other player, resulting in no occlusion.

[0075] In the image **P3** of FIG. **7** captured by a third camera **41**, the target model **TG** does not overlap with any other player, resulting in no occlusion.

[0076] In the image **P4** of FIG. **8** captured by a fourth camera **41**, the target model **TG** partially overlaps with another player **PL2** and the referee **RF1**, resulting in occlusion.

[0077] In the image **P5** of FIG. **9** captured by a fifth camera **41**, the target model **TG** partially overlaps with another player **PL3**, resulting in occlusion.

[0078] Of the five images **P1** to **P5** captured by the five cameras **41**, no occlusion has occurred in two images **P2** and **P3**, while occlusion has occurred in three images **P1**, **P4**, and **P5**. When occlusion occurs frequently in this way, a 3D model of the target model **TG** cannot be generated with high accuracy.

[0079] The 3D model generation unit **12** of the image processing system **1** in FIG. **1** is configured to be able to handle cases where such occlusion occurs frequently and a 3D model cannot be generated with high accuracy by using only the images captured by a plurality of cameras **41**.

### 4. Detailed Configuration Block Diagram of 3D Model Generation Unit

[0080] FIG. **10** is a block diagram illustrating a detailed configuration example of the 3D model generation unit **12**.

[0081] The 3D model generation unit **12** includes an unrigged 3D model generation unit **61** serving as a first 3D model generation unit, a bone information estimation unit **62**, a rigged 3D model generation unit **63** serving as a second 3D model generation unit, and a rigged 3D model estimation

unit **64**.

[0082] Images (captured images) captured by a plurality of cameras **41** installed at the game venue are supplied to the 3D model generation unit **12** via the data acquisition unit **11**. In addition, camera parameters (internal parameters and external parameters) of each of the plurality of cameras **41** are also supplied from the data acquisition unit **11**. The captured images and camera parameters of the cameras **41** are appropriately supplied to each unit of the 3D model generation unit **12**.

[0083] The 3D model generation unit **12** generates, as a target model for which a 3D model is to be generated, a 3D model of each of the subjects appearing in captured images captured by the cameras **41**, such as players and a referee. Data of the generated 3D model of the target model (3D model data) is output to the formatting unit **13** (FIG. **1**).

[0084] The unrigged 3D model generation unit **61** (a first 3D model generation unit) is a 3D model generation unit that generates a 3D model of a target model by using captured images in which no occlusion has occurred. The 3D model of the target model thus generated is referred to as an unrigged 3D model, to distinguish it from a rigged 3D model described below.

[0085] Images **P11** to **P15** in FIGS. **11** to **15** provide examples of captured images from which the unrigged 3D model generation unit **61** generates an unrigged 3D model.

[0086] The images **P11** to **P15** in FIGS. **11** to **15** are images captured during the game by the five cameras **41** at timings different from those of the images **P1** to **P5** in FIGS. **5** to **9**.

[0087] Of the players and the referee appearing in the images **P11** to **P15** in FIGS. **11** to **15**, the player enclosed by a rectangular frame is set as a target model **TG**. It can be seen that no occlusion has occurred on the target model **TG** in the five images **P11** to **P15**. In such a case where no occlusion has occurred on the target model **TG**, for example, a 3D model can be generated in the same manner as the case where a 3D model is generated by capturing images of a target model **TG** in a dedicated studio. For example, the unrigged 3D model generation unit **61** generates a 3D model of the target model **TG** by using a visual hull to sharpen the three-dimensional shape of the target model **TG** using images (silhouette images) from the plurality of cameras **41**, as described above.

[0088] FIG. **16** illustrates a conceptual diagram of data of an unrigged 3D model of the target model **TG** generated by the unrigged 3D model generation unit **61**.

[0089] The 3D model data of the target model **TG** is composed of 3D shape data in a mesh format expressed by a polygon mesh, and texture data which is color information data. In the present embodiment, the texture data generated by the unrigged 3D model generation unit **61** is view independent texture data in which color information remains the same regardless of the viewing direction. Examples of the view independent texture data include UV mapping data in which two-dimensional texture images that are applied to each polygon mesh, which is 3D shape data, is expressed in a UV coordinate system and held.

[0090] Returning to the explanation of FIG. **10**, the unrigged 3D model generation unit **61** supplies the generated 3D model data of the target model (hereinafter referred to as unrigged 3D model data) to the rigged 3D model generation unit **63** and the rigged 3D model estimation unit **64**.

[0091] The bone information estimation unit **62** estimates bone information of the target model from images captured by the plurality of cameras **41** installed at the game venue. The bone information estimation unit **62** also estimates position information of the target model based on the estimated bone information. The bone information of the target model is expressed, for example, for each joint of the human body, by a joint id for identifying the joint, position information (x, y, z) that indicates the three-dimensional position of the joint, and rotation information **R** that indicates the rotation direction of the joint. For example, 17 joints of the human body are set, including the nose, right eye, left eye, right ear, left ear, right shoulder, left shoulder, right elbow, left elbow, right wrist, left wrist, right hip, left hip, right knee, left knee, right ankle, and left ankle. The position information of the target model is defined, for example, by the center coordinates, radius, and height of a circle that defines a cylindrical shape enclosing the target model, calculated



based on the position information (x, y, z) of each joint position of the target model. Based on the captured images that are sequentially supplied as a moving image, the position information and bone information of the target model are sequentially calculated and tracked. The processing of estimating the bone information and position information of the target model can employ any known method. Even when occlusion has occurred in some of the images captured by the plurality of cameras **41**, bone information of the target model can be estimated using captured images in which no occlusion has occurred.

[0092] The bone information estimation unit **62** supplies the position information and bone information of the target model estimated from the captured images as tracking data to the rigged 3D model generation unit **63** and the rigged 3D model estimation unit **64**.

[0093] The rigged 3D model generation unit **63** (a second 3D model generation unit) generates a rigged 3D model of the target model by matching a rigged human body template model to the skeleton and body type (shape) of the target model. The rigged human body template model is a human body model including bone information (rigs) such as bones and joints, and whose body type can be generated parametrically. For the rigged human body template model, a human body parametric model such as SMPL disclosed in NPL 1 can be used. As illustrated in FIG. 17, for a rigged human body template model, in which the shape of the human body is expressed by pose parameters and shape parameters, an offset between each point of the SMPL and the corresponding point of the 3D shape data of the target model is calculated and fitted using the 3D model data of the target model supplied from the unrigged 3D model generation unit **61**. Further, a rigged 3D model of the target model is generated by dividing the target model into meshes to express the shape of the target model in greater detail. The bone information of the rigged human body template model may reflect the bone information of the target model estimated by the bone information estimation unit **62**.

[0094] The rigged 3D model generation unit **63** supplies data of the rigged 3D model of the target model (hereinafter referred to as rigged 3D model data) to the rigged 3D model estimation unit **64**. The rigged 3D model data of the target model only needs to be generated once.

[0095] The rigged 3D model estimation unit **64** is supplied with the unrigged 3D model data of the target model corresponding to the moving image (captured images) captured by each camera **41** from the unrigged 3D model generation unit **61**, and is also supplied with the rigged 3D model data of the target model from the rigged 3D model generation unit **63**. In addition, the rigged 3D model estimation unit **64** is supplied with, as tracking data of the target model, the position information and bone information of the target model corresponding to the moving image (captured images) captured by each camera **41** from the bone information estimation unit **62**.

[0096] The rigged 3D model estimation unit **64** determines whether or not occlusion has occurred on the target model in the captured images from the cameras **41**. Whether or not occlusion has occurred can be determined from the position information of each target model supplied as tracking data from the bone information estimation unit **62** and the camera parameters of each camera **41**. For example, when the target model is viewed from a predetermined camera **41**, if the position information of the target model overlaps with the position information of another subject, it is determined that occlusion has occurred.

[0097] If it is determined that no occlusion has occurred, the rigged 3D model estimation unit **64** selects the unrigged 3D model data of the target model supplied from the unrigged 3D model generation unit **61**, and outputs it as the 3D model data of the target model.

[0098] On the other hand, if it is determined that occlusion has occurred, the rigged 3D model estimation unit **64** generates (estimates) a 3D model of the target model (rigged 3D model) by reflecting the bone information of the target model supplied from the bone information estimation unit **62** in the rigged 3D model of the target model supplied from the rigged 3D model generation unit **63**, as illustrated in FIG. 18. Specifically, the 3D model data of the target model is generated by deforming the rigged 3D model according to the bone information of the target model. The

rigged 3D model estimation unit **64** outputs the generated 3D model data of the target model to a subsequent stage.

#### 5. First 3D Moving Image Generation Processing

[0099] First 3D moving image generation processing by the 3D model generation unit **12** will be described with reference to a flowchart in FIG. **19**. This processing is started, for example, in response to an instruction to generate a 3D moving image through an operation unit (not illustrated). The camera parameters of the cameras **41** installed at the game venue are assumed to be known.

[0100] First, in step **S51**, the 3D model generation unit **12** acquires all the captured images captured by the cameras **41** at a time (hereinafter referred to as a generation time) for a predetermined generation frame of a 3D moving image to be distributed. The 3D moving image to be distributed as a free viewpoint video is to be generated at the same frame rate as moving images captured by the cameras **41**.

[0101] In step **S52**, the 3D model generation unit **12** determines, from one or more subjects in the captured images, a target model for which a 3D model is to be generated.

[0102] In step **S53**, the bone information estimation unit **62** estimates position information and bone information of the target model based on the captured images from the cameras **41**.

Specifically, position information (x, y, z) and rotation information R of each joint of the target model are estimated as bone information, and position information of the target model is estimated as the center coordinates, radius, and height of a cylindrical circle based on the estimated bone information. The estimated position information and bone information of the target model are supplied to the rigged 3D model generation unit **63** and the rigged 3D model estimation unit **64**.

[0103] In step **S54**, the rigged 3D model estimation unit **64** determines whether occlusion has occurred on the target model in captured images from N or more cameras **41** ( $N > 0$ ).

[0104] If it is determined in step **S54** that no occlusion has occurred on the target model in the captured images from N or more cameras **41**, the processing proceeds to step **S55**, and then the unrigged 3D model generation unit **61** generates an unrigged 3D model of the target model at the generation time by using captured images from cameras **41** in which no occlusion has occurred. The generated unrigged 3D model data is supplied to the rigged 3D model estimation unit **64**. The rigged 3D model estimation unit **64** outputs, as 3D model data of the target model, the unrigged 3D model data generated by the unrigged 3D model generation unit **61**.

[0105] On the other hand, if it is determined in step **S54** that occlusion has occurred on the target model in the captured images from N or more cameras **41**, the processing proceeds to step **S56**, and then the unrigged 3D model generation unit **61** searches for a time at which the target model is not occluded by another player from among the captured images at other times that make up the moving image. Next, in step **S57**, the unrigged 3D model generation unit **61** generates an unrigged 3D model of the target model by using the captured images from the cameras **41** at the found time. The generated unrigged 3D model data is supplied to the rigged 3D model generation unit **63**.

[0106] In step **S58**, the rigged 3D model generation unit **63** generates a rigged 3D model of the target model by using the 3D model data of the target model supplied from the unrigged 3D model generation unit **61** for the rigged human body template model. The generated rigged 3D model of the target model is supplied to the rigged 3D model estimation unit **64**.

[0107] In step **S59**, by deforming the rigged 3D model of the target model based on the bone information of the target model at the generation time estimated in step **S53**, the rigged 3D model estimation unit **64** generates a 3D model of the target model at the generation time. The generated 3D model data of the target model is output to a subsequent stage.

[0108] In step **S60**, the 3D model generation unit **12** determines whether all the subjects appearing in the captured images at the generation time have been set as target models. If it is determined in step **S60** that all the subjects have not yet been set as target models, the processing returns to step **S52**, and the above-described processing of steps **S52** to **S60** is performed again. Specifically, a

subject that has not yet been set as a target model in the captured images at the generation time is determined to be the next target model, and 3D model data therefor is generated and output. A rigged 3D model needs to be generated once for one subject (target model), and the processing of steps S56 to S58 for the subject for which a rigged 3D model has been generated from the second time onwards may be omitted.

[0109] On the other hand, if it is determined in step S60 that all the subjects have been set as target models, the processing proceeds to step S61, and then the 3D model generation unit 12 determines whether or not to end the moving image generation. In step S61, if the 3D model generation unit 12 has performed the processing of generating 3D models for captured images at all times that make up a moving image supplied from each camera 41 (the processing of steps S51 to S60), the 3D model generation unit 12 determines to end the moving image generation, and if the 3D model generation unit 12 has not yet performed, the 3D model generation unit 12 determines not to end the moving image generation.

[0110] If it is determined in step S61 that the moving image generation has not yet been ended, the processing returns to step S51, and then the above-described processing of steps S51 to S61 is performed again. Specifically, the processing of generating a 3D model is performed for a captured image at a time when the generation of a 3D model has not yet been ended.

[0111] On the other hand, if it is determined in step S61 that the moving image generation is to be ended, the first 3D moving image generation processing ends.

[0112] The first 3D moving image generation processing is performed as described above. According to the first 3D moving image generation processing, it is determined whether or not occlusion has occurred on the target model in captured images from N or more cameras 41, and if no occlusion has occurred, a 3D model of the target model is generated using a method without using bone information. On the other hand, if occlusion has occurred on the target model, a 3D model of the target model is generated by deforming the rigged 3D model by using bone information.

## 6. Second 3D Moving Image Generation Processing

[0113] The 3D model generation unit 12 can perform second 3D moving image generation processing in FIG. 20 as another different method of generating a 3D moving image.

[0114] The second 3D moving image generation processing by the 3D model generation unit 12 will be described with reference to a flowchart in FIG. 20. This processing is started, for example, in response to an instruction to generate a 3D moving image through an operation unit (not illustrated). The camera parameters of the cameras 41 installed at the game venue are assumed to be known.

[0115] First, in step S81, the 3D model generation unit 12 determines a target model for which a 3D model is to be generated, from among one or more subjects in the captured images from the cameras 41.

[0116] In step S82, the unrigged 3D model generation unit 61 searches for captured images from cameras 41 at a time when the target model is not occluded by another player.

[0117] In step S83, the unrigged 3D model generation unit 61 generates an unrigged 3D model of the target model by using the found captured images from cameras 41. The generated unrigged 3D model data is supplied to the rigged 3D model generation unit 63.

[0118] In step S84, the rigged 3D model generation unit 63 generates a rigged 3D model of the target model by using the 3D model data of the target model supplied from the unrigged 3D model generation unit 61 for the rigged human body template model. The generated rigged 3D model of the target model is supplied to the rigged 3D model estimation unit 64.

[0119] In step S85, the 3D model generation unit 12 determines whether all the subjects in the captured images from the cameras 41 have been set as target models for generating 3D models.

[0120] If it is determined in step S85 that all the subjects have not yet been set as target models, the processing returns to step S81, and then the above-described processing of steps S81 to S85 is

repeated again. Specifically, a subject that has not yet been set as a target model in the captured images is determined to be the next target model, and a rigged 3D model is generated.

[0121] On the other hand, if it is determined in step **S85** that all the subjects have been set as target models, the processing proceeds to step **S86**, and then the 3D model generation unit **12** determines a predetermined generation frame for a 3D moving image to be distributed. As in the first 3D moving image generation processing, the 3D moving image to be distributed as a free viewpoint video is to be generated at the same frame rate as moving images captured by the cameras **41**.

[0122] In step **S87**, the 3D model generation unit **12** acquires all the captured images captured by the cameras **41** at a time (hereinafter referred to as a generation time) for the determined generation frame.

[0123] In step **S88**, the bone information estimation unit **62** estimates position information and bone information of all the subjects appearing in the captured images at the generation time. The estimated position information and bone information of all the subjects are supplied to the rigged 3D model estimation unit **64**.

[0124] In step **S89**, by deforming the rigged 3D models of the subjects based on the bone information estimated in step **S88** for all the subjects appearing in the captured images at the generation time, the rigged 3D model estimation unit **64** generates 3D models of the subjects at the generation time. The generated 3D model data for the subjects is output to a subsequent stage.

[0125] In step **S90**, the 3D model generation unit **12** determines whether or not to end the moving image generation. In step **S90**, the 3D model generation unit **12** determines to end the moving image generation if all the frames that will make up the 3D moving image to be distributed have been generated, and determines not to end the moving image generation if all the frames have not yet been generated.

[0126] If it is determined in step **S90** that the moving image generation has not been ended, the processing returns to step **S86**, and then the above-described processing of steps **S86** to **S90** is performed again. Specifically, the next generation frame for the 3D moving image to be distributed is determined, and processing of generating a 3D model is performed using the captured images at the time for the determined generation frame.

[0127] On the other hand, if it is determined in step **S90** that the moving image generation is to be ended, the second 3D moving image generation processing ends.

[0128] The second 3D moving image generation processing is performed as described above. According to the second 3D moving image generation processing, first, a rigged 3D model of each subject is generated using captured images in which no occlusion has occurred. Then, bone information of each subject appearing in captured images that make up a moving image is tracked (bone information of each subject is sequentially detected), and a 3D model is generated by updating (sequentially deforming) the rigged 3D model based on the tracked bone information.

[0129] The first 3D moving image generation processing and the second 3D moving image generation processing can be appropriately selected and performed according to user settings and the like.

## 7. Conclusion

[0130] The 3D model generation unit **12** includes an unrigged 3D model generation unit **61** (a first 3D model generation unit) that generates an unrigged 3D model that is a 3D model not including bone information of a person, by using captured images captured by the a plurality of cameras **41**, and a rigged 3D model generation unit **63** (a second 3D model generation unit) that generates, based on the bone information of the person, a rigged 3D model that is a 3D model including the bone information of the person. The 3D model generation unit **12** also includes a bone information estimation unit **62** that estimates, based on captured images captured at a predetermined time by the plurality of cameras **41**, bone information of a person appearing in the captured images, and a 3D model estimation unit **64** that, when occlusion has occurred on the person appearing in the captured images at the predetermined time, estimates a 3D model of the person based on the bone

information at the predetermined time.

[0131] Even when occlusion has occurred on the person appearing in the captured images captured by the plurality of cameras **41** at the predetermined time, a 3D model of the person at the predetermined time can be generated with high accuracy by deforming the rigged 3D model of the person based on the bone information at the predetermined time. This makes it possible to generate and distribute high-quality free viewpoint video. Since a 3D model can be generated with high accuracy even when occlusion has occurred, there is no need to increase the number of cameras **41** installed to prevent occlusion from occurring, and high-quality free viewpoint video can be generated with a small number of cameras.

## 8. Configuration Example of Computer

[0132] The series of processing in the image processing system **1** or the 3D model generation unit **12** described above can be performed by hardware or by software. In a case where the series of processing is performed by software, a program that configures the software is installed on a computer. Here, examples of the computer include a computer embedded in dedicated hardware or a general-purpose personal computer capable of executing various functions by installing various programs.

[0133] FIG. **21** is a block diagram illustrating a hardware configuration example of a computer when the computer executes the series of processing performed by the image processing system **1** or the 3D model generation unit **12** by means of a program.

[0134] In the computer, a central processing unit (CPU) **401**, a read-only memory (ROM) **402**, and a random access memory (RAM) **403** are connected to each other by a bus **404**.

[0135] An input/output interface **405** is further connected to the bus **404**. An input unit **406**, an output unit **407**, a storage unit **408**, a communication unit **409**, and a drive **410** are connected to the input/output interface **405**.

[0136] The input unit **406** includes a keyboard, a mouse, a microphone, and the like. The output unit **407** includes a display, a speaker, and the like. The storage unit **408** is a hard disk, non-volatile memory, or the like. The communication unit **409** is a network interface or the like. The drive **410** drives a removable medium **411** such as a magnetic disk, an optical disc, a magneto-optical disk, or a semiconductor memory.

[0137] In the computer configured as described above, for example, the CPU **401** loads a program stored in the storage unit **408** into the RAM **403** via the input/output interface **405** and the bus **404** and executes the program to perform the above-described series of processing.

[0138] The program executed by the computer (the CPU **401**) can be recorded on, for example, the removable medium **411** serving as a package medium for supply. The program can be supplied via a wired or wireless transmission medium such as a local area network, the Internet, or digital satellite broadcasting.

[0139] In the computer, by mounting the removable medium **411** on the drive **410**, it is possible to install the program in the storage unit **408** via the input/output interface **405**. The program can be received by the communication unit **409** via a wired or wireless transfer medium and can be installed in the storage unit **408**. In addition, this program may be installed in advance in the ROM **402** or the storage unit **408**.

[0140] The program executed by a computer may be a program that performs processing chronologically in the order described in the present specification or may be a program that executes processing in parallel or at a necessary timing such as a called time.

## 9. Application Example

[0141] The technology related to the present disclosure can be applied to various products or services.

### 9.1 Production of Content

[0142] For example, new video content may be created by synthesizing a 3D model of the subject generated in the present embodiment with the 3D data managed by another server. Further, for

example, when background data acquired by an imaging device such as Lidar exists, it is possible to also create content in which the subject appears as if the subject is at a place indicated by the background data by combining the 3D model of the subject generated in the present embodiment with the background data. The video content may be three-dimensional video content or may be two-dimensional video content converted into two dimensions. The 3D model of the subject generated in the present embodiment is, for example, a 3D model generated by the 3D model generation unit or a 3D model reconstructed by the rendering unit.

## 9.2 Experience in Virtual Space

[0143] For example, a subject (for example, a performer) generated in the present embodiment can be disposed in a virtual space that is a place at which a user acts as an avatar and performs communication. In this case, the user can act as an avatar and view a live-action subject in the virtual space.

## 9.3 Application to Communication with Remote Place

[0144] For example, it is possible for a user at a remote place to view the 3D model of the subject through a reproduction device at the remote place by transmitting the 3D model of the subject generated by the 3D model generation unit from the transmission unit to a remote place. For example, it is possible for the subject and the user at the remote place to perform communication in real time by transmitting the 3D model of the subject in real time. For example, a case in which the subject is a teacher and the user is a student, or the subject is a doctor and the user is a patient can be assumed.

## 9.4 Others

[0145] For example, it is possible to generate a free viewpoint video of sports or the like on the basis of 3D models of a plurality of subjects generated in the present embodiment, or it is possible for an individual to distribute himself or herself as the 3D model generated in the present embodiment to a distribution platform. As described above, content of the embodiments described in the present specification can be applied to various technologies or services.

[0146] Further, for example, the above-described program may be executed in any device. In that case, the device may have required functional blocks so that the device can obtain required information.

[0147] Further, for example, each step of one flowchart may be performed by one device, or may be shared and executed by a plurality of devices. Further, when a plurality of kinds of processing are included in one step, one device may perform the plurality of kinds of processing, or the plurality of devices may share and perform the plurality of kinds of processing. In other words, it is also possible to perform the plurality of kinds of processing included in one step as processing of a plurality of steps. On the other hand, it is also possible to perform processing described as a plurality of steps collectively as one step.

[0148] Further, for example, in a program that is executed by a computer, processing of steps describing the program may be executed in time series in an order described in the present specification, or may be executed in parallel or individually at a required timing such as when call is made. That is, the processing of the respective steps may be executed in an order different from the above-described order as long as there is no contradiction. Further, the processing of the steps describing this program may be executed in parallel with processing of another program, or may be executed in combination with the processing of the other program.

[0149] Further, for example, a plurality of technologies regarding the present technology can be independently implemented as a single body as long as there is no contradiction. Of course, it is also possible to perform any plurality of the present technologies in combination. For example, it is also possible to implement some or all of the present technologies described in any of the embodiments in combination with some or all of the technologies described in other embodiments. Further, it is also possible to implement some or all of any of the above-described technologies in combination with other technologies not described above.

[0150] The technology of the present disclosure can be configured as follows. [0151] (1)  
[0152] An image processing device including: [0153] a bone information estimation unit that estimates, based on captured images at a predetermined time captured by a plurality of imaging devices, bone information of a person appearing in the captured images; and [0154] a 3D model estimation unit that, when occlusion has occurred on the person appearing in the captured images at the predetermined time, estimates a 3D model of the person based on the bone information at the predetermined time. [0155] (2)  
[0156] The image processing device according to (1), wherein [0157] the bone information estimation unit estimates position information of the person based on the bone information of the person, and [0158] the 3D model estimation unit determines whether occlusion has occurred on the person based on the position information of the person. [0159] (3)  
[0160] The image processing device according to (1) or (2), wherein when occlusion has occurred on the person appearing in captured images from a predetermined number of imaging devices or more, the 3D model estimation unit determines that occlusion has occurred on the person. [0161] (4)  
[0162] The image processing device according to any one of (1) to (3), further including: [0163] a first 3D model generation unit that generates, by using the captured images captured by the plurality of imaging devices, an unrigged 3D model that is a 3D model not including the bone information of the person; and [0164] a second 3D model generation unit that generates, based on the bone information of the person, a rigged 3D model that is a 3D model including the bone information of the person. [0165] (5)  
[0166] The image processing device according to (4), wherein the 3D model estimation unit estimates the 3D model of the person at the predetermined time by deforming the rigged 3D model of the person based on the bone information at the predetermined time. [0167] (6)  
[0168] The image processing device according to (4) or (5), wherein the first 3D model generation unit generates the unrigged 3D model of the person by using captured images in which no occlusion has occurred on the person. [0169] (7)  
[0170] The image processing device according to any one of (4) to (6), wherein [0171] the bone information estimation unit tracks the bone information of the person appearing in the captured images that are sequentially supplied, and [0172] the 3D model estimation unit updates the 3D model of the person based on the tracked bone information. [0173] (8)  
[0174] The image processing device according to any one of (1) to (7), further including a first 3D model generation unit that generates, based on captured images at a predetermined time captured by the plurality of imaging devices, an unrigged 3D model that is a 3D model not including the bone information of the person, [0175] wherein when no occlusion has occurred on the person, the 3D model estimation unit outputs the 3D model of the person generated by the first 3D model generation unit. [0176] (9)  
[0177] An image processing method including: by an image processing device, [0178] estimating, based on captured images at a predetermined time captured by a plurality of imaging devices, bone information of a person appearing in the captured images; and [0179] estimating, when occlusion has occurred on the person appearing in the captured images at the predetermined time, a 3D model of the person based on the bone information at the predetermined time. [0180] (10)  
[0181] A program for causing a computer to execute processing of: [0182] estimating, based on captured images at a predetermined time captured by a plurality of imaging devices, bone information of a person appearing in the captured images; and [0183] estimating, when occlusion has occurred on the person appearing in the captured images at the predetermined time, a 3D model of the person based on the bone information at the predetermined time.

#### REFERENCE SIGNS LIST

[0184] **1** Image processing system [0185] **11** Data acquisition unit [0186] **12** 3D model generation unit [0187] **41** Imaging device (camera) [0188] **61** Unrigged 3D model generation unit (first 3D

model generation unit) [0189] **62** Bone information estimation unit [0190] **63** Rigged 3D model generation unit (second 3D model generation unit) [0191] **64** Rigged 3D model estimation unit [0192] **401** CPU [0193] **402** ROM [0194] **403** RAM [0195] **406** Input unit [0196] **407** Output unit [0197] **408** Storage unit [0198] **409** Communication unit [0199] **410** Drive [0200] **411** Removable medium

## Claims

1. An image processing device comprising: a bone information estimation unit that estimates, based on captured images at a predetermined time captured by a plurality of imaging devices, bone information of a person appearing in the captured images; and a 3D model estimation unit that, when occlusion has occurred on the person appearing in the captured images at the predetermined time, estimates a 3D model of the person based on the bone information at the predetermined time.
2. The image processing device according to claim 1, wherein the bone information estimation unit estimates position information of the person based on the bone information of the person, and the 3D model estimation unit determines whether occlusion has occurred on the person based on the position information of the person.
3. The image processing device according to claim 1, wherein when occlusion has occurred on the person appearing in captured images from a predetermined number of imaging devices or more, the 3D model estimation unit determines that occlusion has occurred on the person.
4. The image processing device according to claim 1, further comprising: a first 3D model generation unit that generates, by using the captured images captured by the plurality of imaging devices, an unrigged 3D model that is a 3D model not including the bone information of the person; and a second 3D model generation unit that generates, based on the bone information of the person, a rigged 3D model that is a 3D model including the bone information of the person.
5. The image processing device according to claim 4, wherein the 3D model estimation unit estimates the 3D model of the person at the predetermined time by deforming the rigged 3D model of the person based on the bone information at the predetermined time.
6. The image processing device according to claim 4, wherein the first 3D model generation unit generates the unrigged 3D model of the person by using captured images in which no occlusion has occurred on the person.
7. The image processing device according to claim 4, wherein the bone information estimation unit tracks the bone information of the person appearing in the captured images that are sequentially supplied, and the 3D model estimation unit updates the 3D model of the person based on the tracked bone information.
8. The image processing device according to claim 1, further comprising a first 3D model generation unit that generates, based on captured images at a predetermined time captured by the plurality of imaging devices, an unrigged 3D model that is a 3D model not including the bone information of the person, wherein when no occlusion has occurred on the person, the 3D model estimation unit outputs the 3D model of the person generated by the first 3D model generation unit.
9. An image processing method comprising: by an image processing device, estimating, based on captured images at a predetermined time captured by a plurality of imaging devices, bone information of a person appearing in the captured images; and estimating, when occlusion has occurred on the person appearing in the captured images at the predetermined time, a 3D model of the person based on the bone information at the predetermined time.
10. A program for causing a computer to execute processing of: estimating, based on captured images at a predetermined time captured by a plurality of imaging devices, bone information of a person appearing in the captured images; and estimating, when occlusion has occurred on the



person appearing in the captured images at the predetermined time, a 3D model of the person based on the bone information at the predetermined time.

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